

Neonicotinoid insecticides negatively affect performance measures of non-target terrestrial arthropods: a meta-analysis

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Abstract. Neonicotinoid insecticides are currently the fastest-growing and most widely used insecticide class worldwide. Valued for their versatility in application, these insecticides may cause deleterious effects in a range of non-target (beneficial) arthropods. However, it remains unclear whether strong patterns exist in terms of their major effects, if broad measures of arthropod performance are negatively affected, or whether different functional groups are equally vulnerable. Here, we present a meta-analysis of 372 observations from 44 field and laboratory studies that describe neonicotinoid effects on 14 arthropod orders across five broad performance measures: abundance, behavior, condition, reproductive success, and survival. Across studies, neonicotinoids negatively affected all performance metrics evaluated; however, magnitude of the effects varied. Arthropod behavior and survival were the most negatively affected and abundance was the least negatively affected. Effects on arthropod functional groups were inconsistent. Pollinator condition, reproductive success, and survival were significantly lower in neonicotinoid treatments compared to untreated controls; whereas, neonicotinoid effects on detritivores were not significant. Although magnitude of arthropod response to neonicotinoids varied among performance measures and functional groups, we documented a consistent negative relationship between exposure to neonicotinoid insecticides in published studies and beneficial arthropod performance.

Key words: abundance; behavior; beneficial insects; condition; meta-analysis; neonicotinoids; non-target; reproductive success; survival; terrestrial.

INTRODUCTION

Beneficial insects and other arthropods found in managed ecosystems (e.g., agricultural) provide numerous services to society including pollination, biological pest control, food resources, and decomposition of plant and animal materials (Van Lenteren and Woets 1988, Kremen and Chaplin-Kramer 2007). In the United States, agricultural production is increased by an estimated US\$8 billion dollars per year through ecosystem services provided by pollinators and insect predators and parasitoids that reduce pest species (Losey and Vaughan 2006). However, global agricultural intensification that maximizes production via inputs (e.g., fertilizers, pesticides), cultivation of marginal land, and reliance on mechanization has decreased landscape diversity and caused *landscape simplification* (Meehan et al. 2011) with implications for beneficial arthropod communities and the services they provide. For example, reductions in habitat provided by field margins (e.g., native plant communities) may decrease crop pollination (Klein et al. 2007) or the predation of agricultural pest species (Tscharntke et al. 2005, Bianchi et al. 2006), which diminish crop yields. Loss of diverse agricultural landscapes is correlative to increased pest pressure, further exacerbating the need for- or use of- insecticide applications (Tilman et al. 2001, Isaacs et al. 2009, Meehan et al. 2011).

Neonicotinoid insecticides are a relatively new, but widely used, class of systemic compounds designed to protect crops from a broad spectrum of agricultural pests (e.g., corn rootworm, flea beetle). However, the broad-spectrum effectiveness of neonicotinoids may cause undesirable direct (e.g., toxicity) and indirect effects (e.g., reduced prey availability) on non-target invertebrates, including beneficial insects (Pisa et al. 2015). Although neonicotinoid seed-treatments can theoretically reduce the quantity of insecticide introduced to the environment compared to spray application, these insecticides have received increased scrutiny as bees may interact with active ingredients via dust, contaminated soil, and/or pollen (Goulson 2013, Godfray et al. 2014). Honey bees and, to a lesser extent, native bee populations (e.g., *Bombus*, *Osmia*) have been the focus of a majority of research evaluating neonicotinoid toxicity toward non-target terrestrial invertebrates (Scott-Dupree et al. 2009, Blacquiere et al. 2012). Far fewer studies have examined potential lethal or sub-lethal effects on known beneficial insects such as predators, parasitoids, and butterflies (Tran et al. 2004, Seagraves and Lundgren 2012, Krischik et al. 2015, Pecenka and Lundgren 2015). It remains unclear if other beneficial terrestrial insects experience similar lethal or sub-lethal effects (e.g., reduced foraging, body condition) due to neonicotinoid exposure.

Neonicotinoids are highly versatile in application with active ingredients (e.g., imidacloprid, clothianidin) used as foliar sprays, soil drenches, and seed-treatments in urban, greenhouse, and agricultural environments (Jeschke et al. 2011, Larson et al. 2013, Krischik et al. 2015). Globally, ~60% of all neonicotinoid applications are soil and seed treatments

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(Jeschke et al. 2011). However, studies have shown that <10% of neonicotinoid active ingredient are accumulated by targeted crops. Subsequently, >90% of active ingredient remains in soil where it may leach to water resources, sorb to soil particles, or degrade (Sur and Stork 2003, Goulson 2013). Neonicotinoids may persist in soils and/or accumulate over time (Goulson 2013, Jones et al. 2014) and, due to high water solubility, neonicotinoids may move from cropped fields to adjacent margin plant communities with potential accumulation by non-target vegetation (e.g., wildflowers; Stewart et al. 2014, Botías et al. 2016). In agroecosystems, non-target beneficial insects may interact with neonicotinoids found in soil, water, plants, and prey depending on application method, contaminant transport processes, and insect biology.

Although numerous studies have recently investigated neonicotinoid effects on performance measures (e.g., reproduction, behavior) of various beneficial insect species, no quantitative synthesis of this growing body of literature currently exists. Furthermore, results from neonicotinoid lab and field studies present highly variable and conflicting results when assessing similar insect response metrics or even the same insect species (e.g., *Orius insidiosus* reproduction; Gontijo et al. 2015, Fernandes et al. 2016). Similar discrepancies exist for other species including honeybees, native bees, butterflies, and earthworms (Blacquiere et al. 2012, Pisa et al. 2015). Many lab experiments use single-species toxicity tests and, although important, it is challenging to extrapolate lab results to field situations (Drobne 1997) or predict responses at higher ecological organization (e.g., communities; Cairns 1983). Discrepancies in the current literature relate to variation in response metrics (e.g., reproduction: fecundity vs. offspring per female), study sample sizes, active ingredients and concentrations tested (e.g., µg/L to mg/L), and controlled lab studies vs. field data. It is essential to understand how neonicotinoid sublethal toxicity endpoints (in field and lab studies) may affect a range of species at levels of the organism (e.g., reproduction, condition) and population (e.g., abundance; Suter et al. 2005).

Previous comprehensive reviews on neonicotinoid insecticides (e.g., Goulson 2013, Chagnon et al. 2015) and their effects on non-target invertebrates (e.g., Blacquiere et al. 2012, Lundin et al. 2015, Pisa et al. 2015) are primarily summarizations of the literature. Although such reviews are important, they typically ignore the influence of sample size and/or statistical significance in comparing results. In contrast, meta-analysis makes it possible to estimate variability in effects across studies (Field and Gillett 2010) while quantitatively analyzing results from numerous studies in an unbiased manner (Gurevitch and Hedges 1993). Meta-analyses can be especially informative when examining often-contested data associated with controversial topics such as the neonicotinoids.

In this paper, we use meta-analysis techniques to synthesize published literature and determine the consistency of neonicotinoid insecticide impacts on specific performance measures of non-target beneficial arthropods. We maintain generality in our findings by gathering existing data from a range of studies (laboratory, semi-field, and field settings) evaluating neonicotinoid use on an equally wide range of

functional groups (e.g., detritivores, pollinators, predators). Our broad focus was to quantitatively evaluate peer-reviewed evidence to determine if neonicotinoid treatments vs. untreated controls had effects on beneficial arthropod abundance, behavior, condition, reproductive success, and survival. We address the following questions: (1) do neonicotinoids affect the five specified measures of arthropod performance, and (2) do the effects of neonicotinoids differ among functional groups (e.g., pollinators, predators)?

METHODS

Literature search

We conducted a literature search (current as of 28 December 2016) using ISI Web of Science, PubMed, and Google Scholar to locate studies evaluating neonicotinoid insecticide effects on non-target beneficial insects. We used the following search phrase: “(neonic* OR clothianidin OR imidacloprid OR thiamethoxam) AND (beneficial OR terrestrial OR non-target OR insect)”. In addition, we used the resultant studies reference lists and existing published reviews (Blacquiere et al. 2012, Chagnon et al. 2015, Pisa et al. 2015) to identify any additional literature missed by search terms. This approach may have missed some papers; however, it provided an unbiased method of collecting relevant literature.

Our primary search located an initial 1,807 publications from which we removed duplicates as well as publications not written in English. Additionally, we excluded book chapters, conference proceedings, theses, or gray literature to insure we could obtain full-text, peer-reviewed articles. We restricted our literature search to terrestrial beneficial arthropods only or those providing a measured ecosystem service (Losey and Vaughan 2006). To that end, we excluded all literature on targeted pest species and aquatic organisms. We did not include data on neonicotinoid impacts to honeybees, as there is an extensive body of literature already published on these beneficial insects (see Cresswell 2011, Blacquiere et al. 2012, Lundin et al. 2015). This procedure yielded 89 studies that underwent a closer inspection beyond the title and abstract. All scientific articles included in our meta-analysis were required to have clearly defined data with the following criteria for inclusion:

- 1) Type of beneficial species. We attempted to gather data from studies evaluating neonicotinoid effects on beneficial arthropods (e.g., predators, detritivores; Losey and Vaughan 2006). We prioritized studies clearly focused on neonicotinoid impacts on a single species; however, this was not always possible based on available literature. We grouped organisms by their functional classification (see *Data extraction*) but some functional group taxa were over-represented in the literature (e.g., pollinators [bumblebee, *Bombus terrestris*], predators [lady beetles, Coccinellidae]). If studies evaluated larger groups of arthropods, we retained the finest level of taxonomic information available (e.g., species > family > order) to eliminate potential taxonomic redundancy presented in the same study. With few exceptions, we retained the most often-reported life stage data (i.e., adults) to

maintain consistency between performance measures. Larval data were used if adult data were unreported. Less-mobile life stages (e.g., larvae) may experience longer exposure to insecticide residues or higher direct toxicity compared to adults, thus results of our meta-analysis likely represent more conservative estimates than if we had focused only on earlier life stages.

- 2) Response variables. Response variables included broad measures of arthropod performance: abundance, behavior, condition, reproductive success, and survival. To investigate possible biases associated with grouping categories of similar variables (foraging, predation) into broad performance measures (e.g., behavior), we evaluated observed effect sizes of individual categories as well as those across functional groups. Although beneficial arthropods may have differing life history traits, the metrics chosen for evaluation were used to identify larger patterns associated with neonicotinoid presence in the environment.
- 3) Type of study. We classified studies into four groups including field, greenhouse, lab, and semi-field studies. Studies that were experimental or observational in nature were included, but only if they included a control (untreated) and treatment scenario. This further enhanced our ability to evaluate results found across study types, which may influence the direction of an effect. Although studies evaluating separate species were preferred, we did collect data from studies measuring effects on larger invertebrate communities when data were reported by order or a finer taxonomic resolution (i.e., family). This approach was most common in field studies measuring abundance.
- 4) Selection of study means. In studies measuring a response variable at different time points (e.g., weekly, monthly, annually), we collected the last mean value provided, rather than select values from early experimental stages or peak observations. We acknowledge this may underestimate potential early life stage or early season effects of neonicotinoid use. We did not calculate overall study means or accept them as part of our evaluation unless that was the only value provided (e.g., seasonal sums).
- 5) Non-independence of studies and/or data. Unless study authors specified separate metrics, only one “count” measure was included from field studies reporting data from visual inspection of insect communities and separate trap data in the same field. We included trap data unless results from visual inspections evaluated a separate taxonomic group not included in the insect traps. This method limited potential issues with visual counts such as observer bias or counts biased against smaller insects. In instances where the same study examined different species or conducted multiple experiments evaluating distinct metrics (e.g., fecundity, percent survival), we considered each of these observations separately. Although this could receive criticism for non-independence of data sources, similar approaches have been used in meta-analyses (Vilà et al. 2011, Gallardo et al. 2016; see Appendix S1). We further tested performance measures separately to avoid potential influence of correlation (Harrison 2011). Last, in studies evaluating multiple

treatments or doses, we collected data from the first five lowest dose group values. Our meta-analysis focused on the more widely used active ingredients of clothianidin, imidacloprid, and thiamethoxam. We included multiple data from studies that evaluated effects of different active ingredients via separate experiments (e.g., imidacloprid, clothianidin; Scholer and Krischik 2014).

It should be noted that neonicotinoid treatments were included regardless of exposure level. Although some treatments are likely to be above concentrations observed in field settings, there is ongoing debate over which concentration values demonstrate “realistic conditions” (Cloyd and Bethke 2011, Wood and Goulson 2017). Similarly, limited data exist regarding impacts on a range of species. With great variation of concentrations detected in the environment and/or used in experimental settings, we chose to include all treatments with sufficient data.

Data extraction

After compiling all studies meeting our established criteria, we determined which categories of beneficial arthropod performance measures were consistently evaluated in the existing literature. Research evaluating neonicotinoid effects on non-target organisms is often highly variable as to the response measures examined (e.g., abundance, fecundity, foraging success). Similar data types and observations (*k*) were grouped under five broad categories of arthropod performance: abundance (150), behavior (24), condition (43), reproductive success (94), and survival (45; Table 1). However, behavior measures that specifically included metrics of time (e.g., foraging; *k* = 14) were analyzed separately due to inverse responses to treatments (with greater amounts of time having a negative impact). Performance measures therefore contain different variables examining a similar outcome (e.g., fecundity or offspring per female both measure reproductive success) or similar variables evaluated by different methods (e.g., abundance is measured by sums per trap and individuals per field). Inclusion of such heterogeneous data allow for quantitative evaluation of patterns through a calculated effect size, a unit-free metric accounting for sample size bias and representing a unique feature of meta-analyses (Vilà et al. 2011). Ultimately, estimating effect sizes from different variables within a category such as the performance measures listed here is fundamental to conducting a robust meta-analysis (Winfree et al. 2009, Vilà et al. 2011, Gallardo et al. 2016).

For each study included in the meta-analysis, we extracted the mean, statistical variation (typically SE), and sample sizes reported for both the untreated control and neonicotinoid treatment(s). Although this constrained the available data, it also greatly reduced the potential for error in our estimates of effect size by limiting interpretation and/or conversion of multiple test statistics (Harrison 2011). In addition, included studies were required to have an obvious untreated control and report replication. All data were extracted from in-text results, tables, or from figures using GetData Graph Digitizer 2.26 (*available online*).⁵ If some of

⁵ <http://getdata-graph-digitizer.com>

TABLE 1. Types of data used to evaluate neonicotinoid effects on overall arthropod performance measures and the proportion of studies that reported these data.

Performance measure and data type	Proportion of studies (%)
Abundance	
Post-treatment counts (e.g., mean per sample after <i>X</i> days)	46
Seasonal sums (e.g., annual mean per field plot)	27
Final sampling mean (e.g., mean of last sampling week)	9
Overall sum (e.g., summary mean after multiple years)	9
Experimental sum (e.g., summary mean per individual experiment)	9
Behavior	
Foraging (e.g., trips per bee, no. workers foraging)	46
Predation (e.g., mean consumption of eggs)	18
Parasitism (e.g., no. ovipositor insertions)	18
Movement (e.g., walk rate)	18
Condition	
Colony mass (e.g., brood stores, wax, bees)	34
Individual mass (e.g., larval or adult weight)	22
Mass of cocoons	22
Size of individual (e.g., body length)	11
Overall biomass	11
Reproductive success	
Brood production (e.g., mean of castes produced)	40
Fecundity (e.g., eggs or offspring per female, no. eggs laid)	36
Cocoon production (e.g., no. produced per individual)	14
Emergence (e.g., mean emergence of males/females)	5
Egg success (e.g., % hatching)	5
Survival	
Adult survival rate (e.g., percent surviving, number alive)	57
Longevity	22
Larval survival (e.g., percent surviving)	14
Brood survival (e.g., no. brood alive)	7

the above criteria were missing (e.g., means, sample sizes), we contacted authors to obtain further information; however, some authors did not respond to our information request or were unwilling to participate. We attempted to have a broad focus by including studies from across the globe, which resulted in the inclusion of research from 12 countries over four continents. Our final database used for the meta-analysis reported here contained 372 observations from 44 peer-reviewed studies (see Appendix S1; Table S1).

We grouped attributes of arthropods (functional group), studies (study type, treatment type, active ingredient), and sub-performance measures (e.g., adult survival, brood survival) into discrete categories to determine neonicotinoid effects on non-target arthropods. All beneficial arthropods were assigned to functional groups based on criteria modified from Wolfenbarger et al. (2008) and Hoekman et al. (2011). Non-target organisms were categorized as one of six functional guilds: detritivores, mixed, omnivores, parasitoids, pollinators, and predators. The mixed guild was

comprised of cosmopolitan individuals or those that could be a mix of predator, omnivore, or herbivore. We based arthropod functional classification decisions on their specified life history stage, detailed species information, or the level reported in the available literature.

Statistical analyses

The effect of neonicotinoid use on arthropod performance measures was examined using standard meta-analytical models. We performed a series of analyses using random-effects models on overall neonicotinoid effects on the five broad performance measures. Second, we performed subsequent meta-regression modelling to evaluate specific sub-categories of arthropod performance (e.g., brood production), and analyzed neonicotinoid effects on six arthropod functional groups to determine if neonicotinoids unequally affect performance metrics. Finally, we used meta-regression (random/mixed-effects) models to evaluate whether study attributes influenced the overall observed effects on non-target arthropod performance. Studies that most-closely matched an individual performance measure category were evaluated together (Harrison 2011).

Effect size calculation

Meta-analyses were based on the difference in effect size between neonicotinoid and control treatments. For each pair of control (untreated) and neonicotinoid (treated) observations extracted (i.e., mean $\pm$ SE), we calculated Hedges' *g* as a measure of effect size (response variable; Viechtbauer 2010). Hedges' *g* is the calculated and bias-corrected standardized mean difference (SMD) or the estimate of the effect size between the treatment and control group, an indication of how much one group may differ from another. Effect sizes are calculated by subtracting the mean of the control from the treatment mean and dividing it by the pooled and weighted standard deviation. Although a unit-free index, Hedges' *g* can range from negative to positive while estimating the size of the impact and its associated direction (positive or negative). An effect size (*g*) was considered significant if the 95% confidence intervals (CI) did not overlap zero.

Using the natural logarithm of the response ratio ($\ln R$; Hedges et al. 1999), we further calculated the mean percentage of change as a response variable to estimate the proportional difference between treatment and control groups. This metric can be easier to interpret than SMD as the units can be back-transformed $[(e^{\ln RR} - 1) \times 100]$ to percentages (Vilà et al. 2011).

We examined categorical comparisons (e.g., sub-categories of performance, functional groups) using *Q* statistics that describe the variation in effect size(s) attributed to differences between the categories evaluated. Using standard random-effects models, we calculated the overall mean effect size (*g*) across the sample of studies containing data on the response measure of interest (e.g., reproductive success). Random-effects models not only account for variation among studies (i.e., sampling error, τ^2 ; Viechtbauer 2010), but are also considered more appropriate for the synthesis of ecological data (Barron et al. 2010, Vilà et al. 2011).

Using the I^2 index, we evaluated whether effect sizes across studies were truly heterogeneous (i.e., variance among effect sizes were greater than sampling error alone). The I^2 index is easily interpreted as a percentage of the total variability in effect sizes due to true heterogeneity or between-studies variability compared to sampling error within studies (e.g., $I^2 = 0$; Wallace et al. 2017).

Meta-regression models

We used meta-regression models to examine variation in effect size attributed to differences in functional groups and sub-category performance measures. The models use functional group and sub-category performance as moderators while adding a random-effects variance component to represent variability within each of the groups (Gallardo et al. 2016). In addition, a priori variables hypothesized to influence neonicotinoid effects on arthropod performance were evaluated. These included variables such as active ingredient, study type (e.g., lab vs. field), and treatment type (e.g., seed treatments vs. foliar applications), which may account for partial heterogeneity in the true effects (Viechtbauer 2010). Total heterogeneity in the data is partitioned into heterogeneity explained by the presence of moderators (Q_M) and unexplained heterogeneity (Q_E) that would be greater than that expected based on sampling error alone. Therefore, a significant Q_E value indicates that significant heterogeneity was unaccounted for by the model, possibly by other moderators not considered (Viechtbauer 2010). When significant differences were identified between categorical levels in random- and mixed-effects models, we completed post-hoc testing in package multcomp (Hothorn et al. 2008) using Tukey tests while controlling for multiple comparisons (i.e., Holm's method).

For our analyses, all models were random- or mixed-effects models fitted using a restricted maximum likelihood estimator (REML) to estimate residual heterogeneity (τ^2), within-study variance (H^2), and between-study variance (Higgins' I^2 ; Viechtbauer 2010). Compared to other variance estimators, the REML provides a balance of unbiasedness and efficiency (Viechtbauer 2005). Hedges' g calculations and all other meta-analyses were conducted using the metafor package (Viechtbauer 2010) in R version 3.2.3 (R Core Team 2015).

We were unable to complete an overall omnibus model with all moderators and hypothesized interactions included due to (1) uneven reporting of study variables in the search literature and (2) potential over-parameterization of models with small representative sample sizes, which can lead to issues of model-fitting or insufficient data to estimate model coefficients (Viechtbauer 2010). Therefore, the meta-analytical models presented here are quantitative evaluations of patterns existing in the literature rather than tests of underlying mechanisms of neonicotinoid effects on arthropods. However, significant results are potentially indicative of areas requiring further investigation.

Publication bias in all data sets was assessed through multiple methods. These included (1) calculation of Rosenthal's fail-safe numbers (Rosenthal 1979), (2) visual inspections of funnel plots and QQ-normal plots of standardized residuals, (3) Egger's test (Egger et al. 1997) and

the subsequent "trim and fill" method to estimate the hypothetical number of missing studies based on funnel plot asymmetry (see Appendix S1: *Publication bias and sensitivity analyses*). However, it is important to note that all methods of calculating potential bias and/or data sensitivity may suffer from misinterpretation or potentially over-interpretation (Egger et al. 1997, Rosenberg 2005, Peters et al. 2007).

RESULTS

Performance measures

Overall, neonicotinoid treatments had a significant negative effect on arthropod performance measures compared to untreated controls using the standardized mean difference (Hedges' g [95% CI] = -0.79 [-0.93 , -0.66], $I^2 = 90.21\%$). When performance metrics were evaluated separately, significant negative effects on arthropod abundance, behavior, condition, reproductive success, and survival were observed (Fig. 1). Behavior was the most negatively affected performance measure ($g = -2.08$ [-2.92 , -1.24], $I^2 = 95.91\%$) compared to that of abundance, which was the least negatively affected ($g = -0.43$ [-0.58 , -0.28], $I^2 = 42.24\%$). However, the trim-and-fill analysis suggested, depending on the specific performance measure, between three and nine studies were missing from the right side of funnel plots (Table 2), indicating a potential bias for publication of more negative results. As all performance measures remained significantly negative even after accounting for trim-and-fill analysis results, we report original model results rather than the revised estimates. Performance of trim-and-fill may be poor when between-study heterogeneity exists or there is true absence of publication bias (Peters et al. 2007). In addition, fail-safe analysis indicated that a large number ($\geq 1,097$) of non-significant observations would be required to reduce the overall significance of our findings (Table 2).

When functional groups were examined individually, there were significant negative effects of neonicotinoids on abundance of arthropods (Table 3). Arthropod abundance was significantly reduced in neonicotinoid treatments for mixed arthropods (Hedges' g [95% CI] = -0.42 [-0.74 , -0.10]), omnivores ($g = -0.89$ [-1.51 , -0.26]), and predators ($g = -0.35$ [-0.57 , -0.12]; Fig. 2). Compared to controls, there were significantly less omnivores ($\sim 72\%$), mixed arthropods ($\sim 32\%$), and predators ($\sim 29\%$) in fields treated with neonicotinoids. However, the effects of neonicotinoid treatments on detritivore and pollinator abundance were not significant, even though the mean effect values were negative. When compared to one another, the magnitude of neonicotinoid effects did not significantly differ between functional groups though sample observations for some functional groups were quite low (e.g., < 5).

Compared to untreated controls, neonicotinoids negatively affected the behavior of predators ($g = -3.46$ [-4.61 , -2.30]), parasitoids, and pollinators (Fig. 2). There were significant differences between functional groups with more negative impacts on predators compared to both parasitoids ($\beta = -2.45$, $z = -2.83$, $P = 0.01$) and pollinators ($\beta = -2.51$, $z = -3.09$, $P = 0.01$). However, as foraging

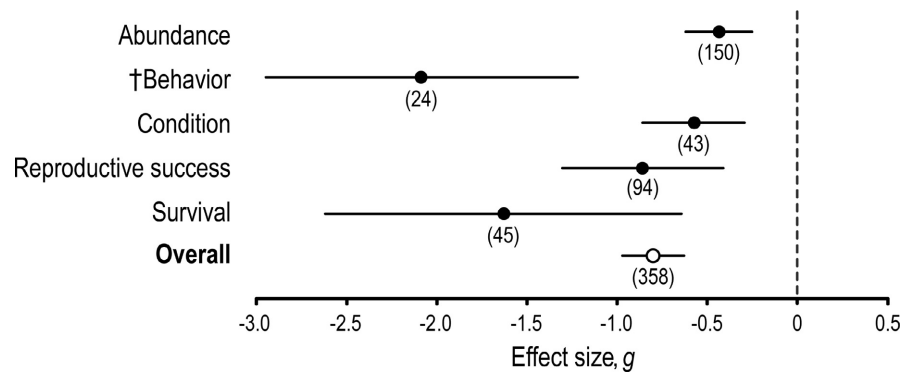


FIG. 1. Mean effect size (Hedges' g) of differences between controls (untreated) and neonicotinoid treatment impacts on arthropod performance measures. Values of $g < 0$ indicate a negative effect whereas values > 0 indicate a positive effect. Bars around means represent 95% confidence intervals (CI). Mean effect sizes are significantly different from zero when the CI does not cross zero. The number of observations included in the effect size calculation for each performance measure are included in parentheses. †Behavior: as foraging metrics were related to time, they were excluded from the calculation of behavior effect size.

TABLE 2. Effects of neonicotinoid treatments on overall arthropod performance, including Q and the associated P values to provide a test for heterogeneity estimated by τ^2 ; I^2 statistic describing the percentage of variation across studies due to heterogeneity rather than sampling error; and sensitivity tests, including trim-and-fill and Rosenthal's fail-safe calculations, presented.

Measure	Q	df	P	τ^2	I^2	Trim-fill†	Fail-safe‡
Abundance	334.14	149	<0.001	0.32	42.24	8	3,987
Behavior	348.01	23	<0.001	3.92	95.91	9	3,090
Condition	190.30	42	<0.001	0.53	89.72	3	1,097
Reproductive success	954.16	93	<0.001	3.85	97.64	—	3,867
Survival	260.89	44	<0.001	2.80	94.23	7	6,188

Note: A dash indicates no value calculated as models did not converge.

†Estimated number of observations potentially missing from analysis of performance measures.

‡Numbers based on calculation of Rosenthal's fail-safe.

TABLE 3. Meta-regression model results evaluating the effect of neonicotinoids on (A) individual functional groups and (B) sub-performance categories (for detailed list, see Fig. 3).

Variables and performance measure	Q_E	df	P	Q_M	df	P	τ^2	I^2
A) Functional group								
Abundance	265.65	134	***	27.95	5	***	0.31	40.96
Behavior	171.89	19	***	39.84	3	***	2.27	93.17
Condition	189.68	41	***	19.79	2	***	0.55	90.24
Reproductive success	920.41	90	***	15.04	4	**	4.37	97.89
Survival	255.36	41	***	43.42	2	***	3.58	95.08
B) Sub-performance								
Behavior	213.91	18	***	36.03	4	***	2.57	93.99
Condition	110.55	38	***	63.94	5	***	0.23	75.33
Reproductive success	896.04	89	***	19.92	5	**	4.15	97.69
Survival	235.17	39	***	49.90	4	***	3.81	95.04

Notes: Q_M and the associated P value provide a test for the effect of the moderators on the mean effect size. Q_E provides a test for residual heterogeneity estimated by τ^2 . The I^2 statistic describes the percentage of variation across studies due to heterogeneity rather than sampling error. Results shown are for functional groups or sub-performance categories that were examined by more than one study.

* $P \leq 0.05$, ** $P \leq 0.01$, *** $P \leq 0.001$.

behavior mainly contained time-related measures (e.g., time to learn behavior, time spent foraging), it was evaluated separately from other measures (e.g., predation). Behavior was moderated by sub-categories of performance ($Q_M = 36.03$, $P = <0.001$; Table 3), and significant reductions in both predation ($g = -3.36$ [$-4.60, -2.12$]) and foraging measures ($g = -0.24$ [$-0.37, -0.11$]) were noted. This analysis

indicated an ~12% decrease in pollinator foraging, but a strong impact on predation with an ~84% reduction in the number of prey consumed by predators exposed to neonicotinoids.

Data evaluating neonicotinoid effects on condition were only available for pollinators and detritivores. Compared to untreated controls, pollinator condition was significantly

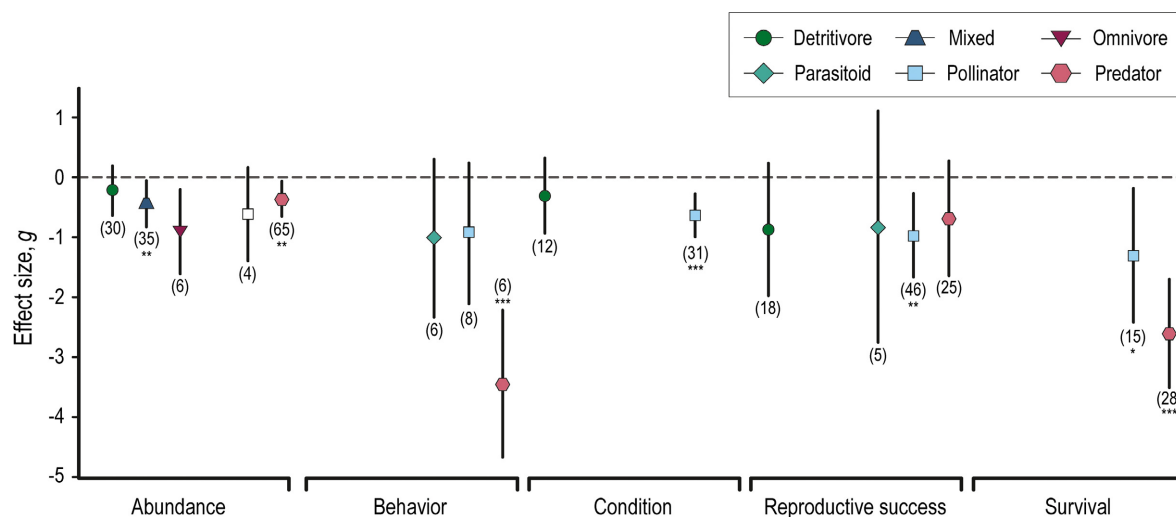


FIG. 2. Effects (Hedges' g) of neonicotinoid treatments on performance measures of non-target arthropod functional groups. Bars around means represent 95% confidence intervals (CI) and number of observations considered in each case are in parentheses. Open symbols indicate fewer than five observations were considered. Spaces indicate no data was available for that functional group. A significant effect of neonicotinoid use is indicated when CI do not overlap zero and significance codes are as follows: *** $P < 0.001$, ** $P < 0.01$, and * $P < 0.05$.

reduced (~25%) when exposed to neonicotinoid treatments ($g = -0.64$ [$-0.93, -0.35$]; Fig. 2), but between-group differences among functional groups were not significant ($\beta = -0.32$, $z = -1.00$, $P = 0.32$). Different effects of neonicotinoid treatments between sub-categories of condition were observed ($Q_M = 63.94$, $P < 0.001$; Table 3). Both colony weight ($g = -1.59$ [$-1.99, -1.17$]) and cocoon weight ($g = -0.63$ [$-1.15, -0.11$]) were significantly reduced, equating to an ~50% reduction in treatment colonies and an ~15% reduction in cocoons measured across studies. However, there were no significant differences in the effects on individual weight of arthropods, overall biomass, and/or size of individuals.

Reproductive success was reduced in response to neonicotinoids for pollinators ($g = -0.97$ [$-1.59, -0.34$]), but not detritivores, parasitoids, or predators (Fig. 2). Pollinators exposed to neonicotinoids had an ~27% reduction in reproductive success compared to controls. The magnitude of neonicotinoid treatment effects on sub-categories of reproductive success was significant ($Q_M = 19.92$, $P = 0.001$). Neonicotinoids significantly reduced egg success ($g = -1.63$ [$-2.67, -0.59$]; Fig. 3) and fecundity ($g = -1.10$ [$-1.89, -0.30$]). Although all other sub-categories were negatively affected, post-hoc comparisons indicated there were no significant differences between effects estimates of reproductive success.

Individually, neonicotinoids more negatively affected predator survival ($g = -2.59$ [$-3.42, -1.76$]) than pollinator survival ($g = -1.31$ [$-2.35, -0.26$]). Across observations, predator survival was approximately 65% less in treatments compared to controls. Although the size of the effect differed between functional groups, they were not statistically different from one another ($\beta = -1.29$, $z = -1.89$, $P = 0.058$). All individual measures of survival were negatively affected by neonicotinoid treatments; however, brood survival was not significantly different between treatments and controls (Fig. 3). Magnitude of the effect also differed between survival measures, with adult survival time

significantly less than differences in brood survival ($\beta = -2.78$, $z = -3.31$, $P = 0.006$).

Influence of study variables on neonicotinoid effects

Study variables influenced the overall effects of neonicotinoids on arthropod performance measures (Table 4). Evaluated by separate meta-regression models, active ingredient had a significant negative influence on reproductive success ($Q_M = 10.18$, $P = 0.006$, $R^2 = 9.79\%$), but multiple comparisons indicated no significant differences between active ingredients. In comparison, active ingredient also influenced arthropod survival ($Q_M = 13.02$, $P = 0.002$) accounting for 26% of overall variation in effects. Survival was diminished for arthropods exposed to imidacloprid ($\beta = -1.97$, $z = -2.97$, $P = 0.003$) compared to other active ingredients. However, post-hoc comparisons indicated standardized mean differences between treatments and controls were much less in studies that used thiamethoxam compared to clothianidin ($\beta = -1.93$, $z = -2.61$, $P = 0.03$) and no statistical difference compared to imidacloprid ($\beta = -1.48$, $z = -1.22$, $P = 0.44$).

Application type influenced the negative effect of neonicotinoids on arthropod abundance ($Q_M = 18.08$, $P = 0.001$) and survival ($Q_M = 17.35$, $P = 0.002$). Arthropod abundance was significantly less in studies using neonicotinoid foliar applications compared to granular/wettable powders ($\beta = -1.72$, $z = -2.89$, $P = 0.03$), lab compounds ($\beta = -1.95$, $z = -2.73$, $P = 0.04$), seed-treatments ($\beta = -2.06$, $z = -3.47$, $P = 0.005$), and soil injections ($\beta = -2.40$, $z = -3.69$, $P = 0.002$). In contrast, arthropod survival was significantly reduced in studies using seed treatments compared to granular/wettable powders ($\beta = -2.32$, $z = -3.24$, $P = 0.01$) or soil injections ($\beta = -3.11$, $z = -3.04$, $P = 0.02$). Although significant, application type explained less than 2% of the variation in either model.

The type of study (e.g., lab vs. semi-field) explained some variation in the negative effects on behavior ($R^2 = 27.9\%$)

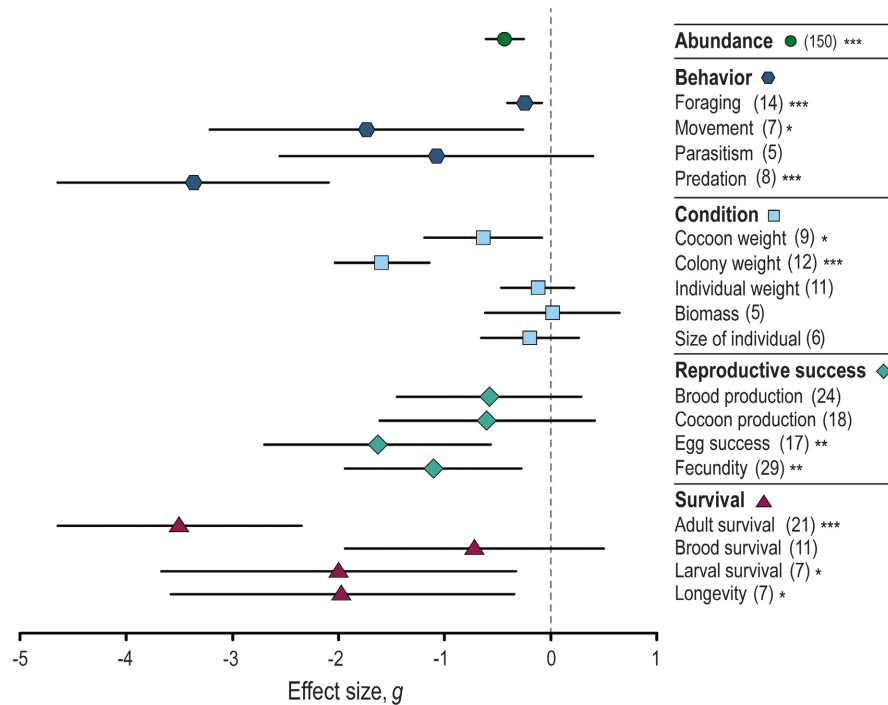


FIG. 3. Effects (Hedges' g) of neonicotinoid treatments on specific ecological measures categorized by overall arthropod performance (boldface type) and sub-category measures. Only sub-categories with sufficient data (more than five observations) are illustrated with shapes indicating their associated performance measure categories. Bars around means represent 95% confidence intervals (CI) and the number of observations considered in each case are in parentheses. A significant effect of neonicotinoid use is indicated when CI do not overlap zero and significance codes are as follows: *** $P < 0.001$, ** $P < 0.01$, and * $P < 0.05$.

TABLE 4. Meta-regressions of (A) application type (lab compound, foliar sprays, powder/granular, seed, or soil treatments), (B) study type (field, greenhouse, lab, or semi-field), and (C) active ingredient (clothianidin, imidacloprid, or thiamethoxam) and their individual influence on reduction in arthropod performance by neonicotinoids.

Variable and performance measure†	Q_E	df	P	Q_M	df	P	τ^2	I^2
A) Application type								
Abundance	255.42	134	***	18.08	4	**	0.30	41.34
Behavior	225.32	21	***	4.77	2	ns	3.22	94.82
Condition	175.97	39	***	4.07	3	ns	0.51	89.97
Reproductive success	940.59	89	***	0.37	4	ns	4.50	97.96
Survival	231.45	40	***	17.35	4	**	2.99	93.40
B) Study type								
Behavior	249.27	22	***	7.97	1	**	2.83	94.27
Condition	114.31	39	***	39.38	3	***	0.22	76.77
Reproductive success	910.77	90	***	0.25	3	ns	4.28	97.90
Survival	252.15	41	***	7.44	3	ns	3.63	95.25
C) Active ingredient								
Behavior	322.42	22	***	2.17	1	ns	3.84	95.77
Condition	161.45	40	***	3.26	2	ns	0.50	87.92
Reproductive success	905.97	91	***	10.18	2	**	3.48	97.34
Survival	214.99	42	***	13.02	2	**	2.07	91.69

Notes: Q_M and the associated P value provide a test for the effect of the moderators on the mean effect size. Q_E provides a test for residual heterogeneity estimated by τ^2 . The I^2 statistic describes the percentage of variation across studies due to heterogeneity rather than sampling error. ns, not significant.

†Active ingredient and study type were excluded for abundance as all observations were field studies using imidacloprid.

* $P \leq 0.05$, ** $P \leq 0.01$, *** $P \leq 0.001$.

and the majority of variation regarding condition ($R^2 = 59.2\%$). Behavior studies included in this analysis were limited to lab and semi-field conditions; however, behavioral activity in lab studies was significantly reduced compared to

those studies conducted in semi-field conditions ($\beta = -4.99$, $z = -4.46$, $P < 0.001$). Arthropod condition was significantly less in greenhouse studies compared to laboratories ($\beta = -1.68$, $z = -4.60$, $P < 0.001$), semi-field ($\beta = -1.74$,

$z = -4.74$, $P < 0.001$) or field situations ($\beta = -1.57$, $z = -3.72$, $P < 0.001$). Exposure ranges, however, often differed among study types.

DISCUSSION

Our analysis provides quantitative evidence that neonicotinoid insecticides consistently, negatively affect non-target arthropod performance across functional groups. However, the magnitude of negative effect varied with the most evidenced effects exerted on arthropod behavior and survival. The most substantial effects on 17 specific response measures were reduced predation rates, movement, colony mass, egg success, and survival rates in adult arthropods. However, no single functional group was negatively affected by neonicotinoids across all performance measures evaluated and nine sub-category measures were unaffected by neonicotinoid treatments. Pollinator condition, reproductive success, and survival were consistently lower in neonicotinoid treatments compared to controls. However, the opposite was true for detritivores where, across studies, effects of neonicotinoid treatments did not significantly differ compared to controls. Although these results might indicate that neonicotinoids are not detrimental to some functional groups or metrics of performance, more studies are required.

In this quantitative synthesis, we were restricted to testing broad categories of functional groups and performance for which we had sufficient data. We acknowledge that our search terms may have limited our scope with subsequent data missed for inclusion in this synthesis. Conversely, numerous (and potentially valuable) studies excluded basic information on means, sample sizes, measures of variation or did not accurately report test statistics or units—a known problem for inclusion in meta-analyses or evaluating broader impacts (Gerstner et al. 2017). As our findings were similar across metrics evaluated, this gives strength to our quantitative approach suggesting a wide variety of non-target functional groups are likely to be negatively impacted by use of neonicotinoids.

Perhaps the most unexpected result of our study is that neonicotinoids negatively influenced all five arthropod performance metrics evaluated. The effects of neonicotinoid use were most evidenced as follows: behavior > survival > reproductive success > condition > abundance. It should be noted that Cohen's d and Hedges' g are interpreted in a similar way such that a small effect is 0.2, a medium effect is 0.5, a large effect is 0.8, and a very large effect >1 (Rice and Harris 2005). Effect sizes presented herein often ranged between medium to large, with some effects on functional groups and/or performance measures illustrating potential very large impacts when exposed to neonicotinoids. Our results demonstrate that neonicotinoids negatively impact the majority of arthropod behaviors evaluated, and much of the negative effect on behavior may be influenced by decreases in predation rates. Predators exposed to neonicotinoids experienced substantial decreases (~84%) in the ability to locate and attack prey and/or consume eggs and other organisms. Depending on the agrochemical of interest, pesticides may interfere with arthropod predators via repellency or decreased ability to locate prey due to

reduced olfactory capacity (Desneux et al. 2007). Neonicotinoids can reduce predator density (Douglas et al. 2015) and suppress predation on pest populations regardless of active ingredient or season of application (Peck and Olmstead 2010). Inability to obtain prey may also explain why predators experienced the greatest decreases in survival, which may have a more negative effect for long-term population persistence in various arthropod predators.

In contrast, neonicotinoid treatments had a small but negative effect ($g = -0.24$) on foraging behavior. Though some studies have evaluated neonicotinoid use on foraging behavior and found negative effects (Feltham et al. 2014, Stanley et al. 2016), others have found no effect of neonicotinoids on individual foraging or colonies (Morandin and Winston 2003). Similar foraging observations across studies were not universally reported and included metrics such as time spent foraging and/or amount of pollen collected upon return to the colony (Morandin and Winston 2003, Stanley and Raine 2016, Stanley et al. 2016). Foraging behavior may be learned faster in bees exposed to neonicotinoids but time spent foraging is longer (Stanley and Raine 2016), which could be the result of bees preferring neonicotinoids in their food sources in spite of the potential for harm (Kessler et al. 2015). However, an increase in time spent foraging could lead to decreased pollinator survival in social bees due to greater losses of workers, impaired orientation, and reduced colony development or pollen gathered for the colony (Gill et al. 2012, Whitehorn et al. 2012). Indeed, we found pollinator survival was significantly reduced in neonicotinoid treatments compared to controls. As there is a scarcity of data evaluating neonicotinoid impacts to native solitary bees, there is insufficient evidence to conclude whether foraging behavior could be negatively impacted across the majority of global bee populations.

As neonicotinoids are designed to kill insects, it is unsurprising that they can affect survival of numerous beneficial arthropods (Pisa et al. 2015). Early life stages and/or organisms with more life stages are generally the most sensitive to pesticides (Stark et al. 2004, Pisa et al. 2015). Our results indicate that arthropod survival was negatively affected at all life history stages examined, indicating the ability of neonicotinoids to cause potential harm beyond the least mobile life stage (Larson et al. 2013, Krischik et al. 2015). It is likely that effects on other performance metrics would appear less detrimental by comparison as loss of individuals could increase the likelihood for declines in condition, reproductive success, and ultimately ecosystem service production (Köhler and Triebkorn 2013). As the effect on survival was "very large" for both predators ($g = -2.59$) and pollinators ($g = -1.31$), our results suggest potentially greater reductions of ecosystem service provisioning by a range of functional groups. However, the lethality of neonicotinoids varies greatly depending on the route of exposure, species, concentrations, and active ingredient used (Scott-Dupree et al. 2009, Fernandes et al. 2016, Spurgeon et al. 2016).

Routes of exposure, concentrations, and experimental conditions often vary greatly between field and lab studies (Larson et al. 2013, Scholer and Krischik 2014, Rundlöf et al. 2015). There was a strong influence of study type ($R^2 = 59.2\%$) on effects to arthropod behavior with negative

effects on behavior most evidenced in lab studies compared to semi-field conditions. However, exposure concentrations across studies varied depending on prescribed application rates and/or doses (ppb to ppm) examined in the trials (Tran et al. 2004, Stanley et al. 2016). Without baseline data from field studies, it is difficult to compare results of lab studies and make conclusions on potential lethal and sublethal impacts of neonicotinoids beyond a select number of species (Wood and Goulson 2017). As semi-field and field studies are unable to control for environmental conditions (e.g., climate, exposure to multiple pesticides), discerning neonicotinoid effects on non-target arthropods may be difficult due to external pressures (Goulson 2013, Thompson et al. 2013).

Few studies have examined arthropod condition and among these, most focused on detritivores (e.g., earthworms) and pollinators (e.g., bumblebees). Measures such as individual insect size, biomass, or colony mass are typically used as a metrics of organismal fitness. Among condition measures evaluated in our meta-analysis, cocoon mass and colony mass were significantly reduced for non-target beneficial arthropods exposed to neonicotinoid treatments when compared to untreated controls. In particular, this trend exhibited a greater impact on the effect on colony weight ($g = -1.59$). Beyond honeybees, most studies have examined *Bombus impatiens* or *B. terrestris*, with far fewer examining condition across a range of other pollinators (e.g., butterflies; Goulson 2013, Pecenka and Lundgren 2015) or other beneficial insects (e.g., natural predators). However, studies have shown that sub-lethal impacts of neonicotinoids, such as reduced queen or worker production, may lower overall bumblebee colony success (Whitehorn et al. 2012, Larson et al. 2013, Rundlöf et al. 2015). It remains unclear how metrics of condition for a range of other beneficial arthropods are likely to be affected by exposure to neonicotinoid use; more laboratory and field studies are needed.

All sub-category measures of reproductive success were negatively affected by neonicotinoid treatments with significant effects on both fecundity and egg success (i.e., percentage of eggs yielding viable neonates). Across a range of taxa (e.g., Family: Apidae, Anthoridae, Cantharidae, Coccinellidae) and guilds of study organisms, only effects on the pollinator guild were significant, despite negative effects of neonicotinoid use across all functional groups. Under experimental manipulation, native bees (e.g., *B. terrestris*) and pollinating insects (e.g., soldier beetles, Family: Cantharidae) may experience reduced fecundity when exposed to neonicotinoids (Elston et al. 2013, Fernandes et al. 2016). However, organismal response to neonicotinoids is influenced by sensitivity to active ingredients and exposure concentrations (Whitehorn et al. 2012, Gontijo et al. 2014, Scholer and Krischik 2014, Fernandes et al. 2016). In our analysis, active ingredient explained <10% of total heterogeneity between studies, and although significant, there were no differences between effects of active ingredients. Concentrations used in experiments or detected in the environment were highly variable with insufficient reporting to calculate common units, application rates or doses, thus precluding comparing concentration effects on reproductive success. As the majority of studies evaluated reproductive success within

a controlled environmental setting, it is difficult to speculate whether this trend would be similar for a range of arthropods in their natural habitats. Few studies have assessed neonicotinoid effects on pollinators in true field settings (Cutler and Scott-Dupree 2014, Rundlöf et al. 2015, Peters et al. 2016) or evaluated effects beyond a single representative species (Baron et al. 2017). However, a more recent study indicated neonicotinoid exposure from non-target sources reduced overwintering success and colony reproduction in wild bees (Woodcock et al. 2017).

Arthropod abundance was significantly reduced in treated vs. untreated conditions, but had the lowest magnitude in overall effect size relative to other performance measures. These results suggest there may be stronger effects at the individual organism and/or species level that are masked at the level of overall abundance by fluctuations between groups or by other compensatory mechanisms. Although there have been studies demonstrating neonicotinoid effects at the population level, these are likely species specific (Godfray et al. 2014, Pisa et al. 2015) and too few data exist on many beneficial arthropods beyond a select group of pollinators, predators, and detritivores to make universal inferences between abundance and neonicotinoids. To date, experimental field studies have primarily measured abundance of non-target arthropods following neonicotinoid application (Knoepf et al. 2012, Seagraves and Lundgren 2012). However, with few exceptions (e.g., Knoepf et al. 2012), associated concentration data were not quantified, making potential effects of actual environmental exposure more ambiguous. Predators, omnivores, and mixed arthropod abundance were negatively affected by neonicotinoid treatments, which may indicate negative effects on food webs (Douglas et al. 2015). To our knowledge, virtually no data are available on pollinator or parasitoid abundance. Consequently, it is difficult to identify similar negative trends on non-target communities such as solitary bees, butterflies, or parasitoids. It should be noted that a 2016 multi-species occupancy analysis of native bee populations in the UK found evidence of population losses in response to neonicotinoid seed treatment use on oilseed rape (Woodcock et al. 2016). Although many studies have evaluated the use of seed treatments or granular applications on arthropod abundance, far fewer have studied the effect of foliar applications. We found that abundance was significantly reduced in studies using neonicotinoids as a foliar spray compared to other applications (e.g., seed treatments, soil injections). Foliar sprays may increase the bioavailability or non-target organism contact of the active ingredient depending on the target crop; thus, they are considered more harmful to natural predators than other application methods (Cloyd and Bethke, 2011). However, interpretation of the meta-analysis results requires caution, as only two studies included use of a foliar spray.

Neonicotinoid concentrations

Depending on the specific study and/or performance measure evaluated, treatment concentrations could vary by orders of magnitude. We were unable to standardize and/or evaluate neonicotinoid concentration in our meta-analysis due to inconsistencies in reporting; however, “realistic

conditions" are often disputed in the literature (Cloyd and Bethke 2011, Wood and Goulson 2017) and levels of exposure for many species are poorly understood (Pisa et al. 2015). Approximately 70% of included study observations indicated either neonicotinoid treatments followed recommended label application rates and/or occurred at "field realistic" values. We acknowledge that by including some data at potentially unrealistic concentrations, this may have influenced our study outcomes. Therefore, some caution is advised when evaluating the effects and resultant figures presented herein. However, results of our sensitivity analyses (i.e., leave-one-out) indicated that, although some studies using higher concentrations increased the magnitude of effect, the direction of effect did not change even if those same studies were removed. Importantly, non-target arthropods may be impacted at a range of concentrations with a number of studies demonstrating negative impacts on arthropod performance at (or below) recommended label or application rates (e.g., abundance; Peck 2009, Seagraves and Lundgren 2012) or levels measured in the environment (e.g., reproductive success; Sandrock et al. 2014, Rundlöf et al. 2015).

CONCLUSIONS

Our meta-analysis demonstrated that neonicotinoid treatments significantly reduce terrestrial arthropod performance across broad ecological metrics including abundance, behavior, condition, reproductive success, and survival even though data limitations and differences in functional groups exist in the published literature. All observed effects were significantly negative, with only the magnitude of the effect differing among measures. Because these effects may be highly context dependent, more studies that (1) evaluate similar metrics, (2) repeat observations across time and on larger scales, and (3) assess neonicotinoid effects across a larger spectrum of organisms are needed. Studies on Hymenoptera (excluding parasitoids) and Coleoptera were over-represented in the literature compared to Diptera or Lepidoptera and this bias may limit our ability to make broad or general conclusions about the effects of neonicotinoids across a broad range of terrestrial taxa. A greater diversity of studies would greatly facilitate the evaluation of potential neonicotinoid effects from the individual to the population. In particular, more experimental field studies would greatly increase our knowledge of neonicotinoids as one of multiple environmental stressors, as controlled lab experiments may be over- or under-estimating effects. Current understanding of neonicotinoid effects is restricted to relatively few non-target terrestrial species with most literature focused on honeybees (Cresswell 2011, Lundin et al. 2015) and effects on a broader range of non-target species are still mostly unknown. Predicting the potential consequences of neonicotinoid use on ecosystems will require not only studies that expose beneficial insects in the laboratory, but also improved understanding of environmental concentrations and likelihood of exposure of beneficial insects in nature. Understanding this information gap is critical for future management decisions that may compare neonicotinoid use against alternative management strategies (e.g., integrated pest management,

IPM). Regardless, results of our meta-analysis indicate that across studies, neonicotinoids are negatively affecting non-target terrestrial arthropods at the doses being tested and reported, information we hope provides areas for further exploration and hypothesis testing.

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A. Main conducted the literature search, performed the meta-analysis, and wrote the first draft of the manuscript. E. Webb, K. Goynes, and D. Mengel provided intellectual guidance and contributed to initial versions of the manuscript. All authors contributed substantially to revisions.

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SUPPORTING INFORMATION

Additional supporting information may be found online at: <http://onlinelibrary.wiley.com/doi/10.1002/eap.1723/full>

DATA AVAILABILITY

Data available from Figshare: <https://doi.org/10.6084/m9.figshare.5999684>